

Paying Off: Evaluating Credit Card Minimum Payment Requirements

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The views presented are those of the authors and do not necessarily reflect those of the Bank of Canada

Key Challenge of Consumer Financial Protection

“To help consumers who are making mistakes in their credit usage without imposing too great a cost on consumers who are benefiting from the availability of short-term credit”
– John Y. Campbell (2016)

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One natural policy option is disclosure or nudges

- Attractive because it does not constrain choices (may expect trivial costs, so net benefit)
- Unfortunately, in credit card domain, these are ineffective at changing long-run behaviors (Agarwal et al., 2015; Seira et al., 2017; Adams et al., 2022; Guttman-Kenney et al., 2025; Batista et al., 2026; Guttman-Kenney, 2026; Han & Yin, 2026; Schwartz, 2026)

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When disclosures and nudges are ineffective, one option is “**hard paternalism**”

- Policies restricting choice

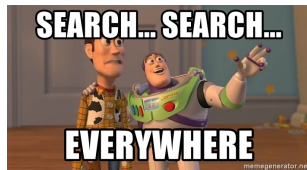
A policy's effects depends on estimating the costs and benefits across consumer groups:

- Benefits? Reduce debt of “unconstrained” consumers who overborrow
- Costs? Inefficient default, credit rationing for liquidity-constrained consumers

Hard Paternalistic Policy To Reduce Credit Card Debt?

Mandating Tighter (Higher) Credit Card Minimum Payments

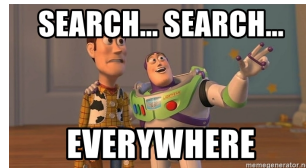
- Want policy forcing credit card lenders involuntarily to increase minimum payment
- Policy variation does not exist in USA



Hard Paternalistic Policy To Reduce Credit Card Debt?

Mandating Tighter (Higher) Credit Card Minimum Payments

- Want policy forcing credit card lenders involuntarily to increase minimum payment
- Policy variation does not exist in USA
- **Canada: 90% of consumers hold credit cards**



Research Question

What are the trade-offs of a market-wide credit card policy that restricts consumer choices?

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Setting: Credit card minimum payment mandate in Quebec (Rest of Canada unaffected)

Data & Methodologies:

- Bank of Canada TransUnion consumer credit reporting data + Mintel card offers
- Synthetic Difference-in-Differences
- Theory model to understand supply responses
- Consumer welfare evaluation using sufficient statistics approach after accounting for supply-side responses

Research Question

What are the trade-offs of a market-wide credit card policy that restricts consumer choices?

Key Results:

- Increases minimum payments: ↑ 75% or \$89 per month after six years
- Substantial decreases in credit card revolving debt: ↓ 26% or \$618 after six years
- Increases in delinquencies but not defaults
- Reduction in credit access
- Consumer welfare rises unless lost access worth over \$16 to \$54 per Quebec consumer (compared to over \$56 to \$148 without supply correction)

Contribute To Household Finance, Behavioral & Public Economics Literatures

- Estimate trade-off of hard paternalistic policy in credit market
 - Allcott et al. (2022) payday lending study is closest prior work (no supply-side response)
 - Prior work shows interest rate caps are a blunt tool
(Cuesta & Sepulveda, 2021; Melzer, 2021; Chakrabarti et al., 2025; Cherry, 2026)
 - We study heterogeneity to consider if minimum payments can be a better-targeted policy
- Consumer financial protection policy that is effective at reducing long-run credit card debt
 - Contrast to literature on ineffectiveness of disclosure and nudges in this domain

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- Consumer financial protection policy that is effective at reducing long-run credit card debt
 - Contrast to literature on ineffectiveness of disclosure and nudges in this domain
- Welfare effects of economy-wide minimum payment policy accounting for supply responses
 - Prior research studied lenders *voluntarily* changing minimum payments (d'Astous & Shore, 2017; Keys & Wang, 2019; Medina & Negrin, 2022; Castellanos et al., 2026), & Turkish policy effectively preventing using credit cards for borrowing (Agarwal et al., 2023)
 - Prior work does not observe a consumer's credit card portfolio of revolving debt
- Policy-relevant variation: From below US level (1% balance + interest + fees) to above

1. Quebec Policy
2. Data & Methodology
3. Consumers
4. Credit Supply
5. Welfare

Quebec Policy

Quebec Is A Laboratory For Hard Paternalism Aiming To ↓ Credit Card Debt

Quebec Office de la Protection du Consommateur:

*“These changes are part of the measures adopted by the government
to prevent indebtedness”*

*“It is advantageous to pay the balance of your credit card each month,
because no credit charges are then applicable”*

Quebec rolls out new credit card rules aimed at lowering high household debt

By Reuters Staff

2 MIN READ



OTTAWA, Aug 2 (Reuters) - The Canadian province of Quebec has become the first in the country to require minimum credit card payments as legislators try to tackle rising household debt levels.

Quebec introduces credit card regulations aimed at curbing consumer debt

CTVNews.ca Staff



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Published Thursday, August 1, 2019 4:38PM EDT

Quebec Minimum Payment Policy (“Bill 134”)

No credit card minimum payment requirements:

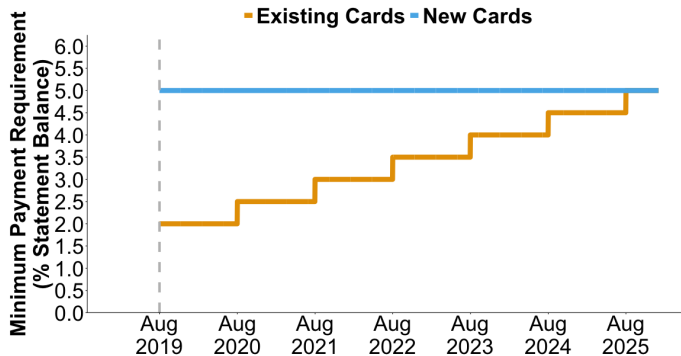
- Pre-August 2019: All Canadian provinces
- Post-August 2019: All Canadian provinces **except Quebec**

Quebec Minimum Payment Policy (“Bill 134”)

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- Post-August 2019: All Canadian provinces **except Quebec**

Quebec 2019 credit card minimum payment policy phases:



Data & Methodology

Consumer Credit Reporting Data

Bank of Canada's anonymized TransUnion consumer credit reporting data

- 2018 to 2026
 - In August 2019, 24.4 million consumers in Canada and 5.8 million (24%) in Quebec
- Monthly, tradeline-level information on each credit card held in Canada
- We observe:
Statement balances, credit limits, minimum payments, actual payments, delinquency status

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Mintel CompereMedia Data

- Credit card offers received via mail/email by 8,000 Canadian households
- Includes contract terms (e.g., APRs, credit limits)
- Measure of supply (e.g., Han, Keys, & Li, 2018; Ru & Schoar, 2020; Grodzicki, 2022)

Synthetic Difference-in-Differences (SDID) Methodology

Arkhangelsky, Athey, Hirshberg, Imbens & Wager (2021 AER); Arkhangelsky & Imbens (2024):
– SDID outperforms both synthetic controls & difference-in-differences approaches

Follow Arkhangelsky et al. (2021) estimating:

$$(\hat{\delta}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \underset{\delta, \mu, \alpha, \beta}{\operatorname{argmin}} \left\{ \sum_{p=1}^9 \sum_{t=-18}^{78} (y_{p,t} - \mu - \alpha_p - \beta_t - \delta(\mathbb{I}\{t \geq 0\} \times \mathbb{I}\{p = \text{QUEBEC}\}))^2 \hat{\omega}_p \hat{\lambda}_t \right\} \quad (2)$$

- Outcomes in Quebec (Q) are compared to “Synthetic Quebec” (SQ) of 8 provinces
- Weights using statement balances $t = -18$ to $t = -13$ to allow for anticipation effects

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- Weights using statement balances $t = -18$ to $t = -13$ to allow for anticipation effects
- Use same weights for all outcomes (weights range from 0.11 to 0.14)
- Standard errors using placebo variance estimation to account for single treated cluster
- Show dynamic monthly effects (δ_τ) relative to pre-policy baseline ($t = -18$ to $t = -13$):
- Also show pooled estimates for each phase of the policy (δ_j^{PHASE}) to increase power
- Robust to difference-in-differences approach

Show Effects Over Each Policy Phase & Across Groups of Consumers

Policy Phases:

I: 2.0% (August 2019)

II: 2.5% (August 2020)

III: 3.0% (August 2021)

IV: 3.5% (August 2022)

V: 4.0% (August 2023)

VI: 4.5% (August 2024)

VII: 5.0% (August 2025)

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VII: 5.0% (August 2025)

Heterogeneity by Consumers Based on Pre-Policy Behaviors:

- **Transacting Status:** Transactor, Revolver
- **Credit Score:** Subprime, Near Prime, Prime, Prime Plus, Superprime
- **Payer Type:** Exact Min., Near Min., Mixed, Full Payer (Keys & Wang, 2019)
- **Balance-Matching** (Gathergood, Mahoney, Stewart, & Weber, 2019)

Consumers

How Lenders Implemented First Policy Phase In August 2019?

Non-binding, Bind, Bind And Go Beyond

Lender	Existing Cards Opened Before August 2019			
	July 2019		August 2019	
	Non-Quebec	Quebec	Non-Quebec	Quebec
A,B	\$10 + interest + fees		max {2.0% statement balance, \$10}	
C,D,E	\$10 + interest + fees		max {2.5% statement balance, \$10}	
F	\$10 + interest + fees		max {3.0% statement balance, \$10}	
G	max {1.0% statement balance, \$10}		max {3.0% statement balance, \$10}	
H,I	max {2.0% statement balance, \$10}		max {2.0% statement balance, \$10}	
J	max {2.5% statement balance, \$10}		max {2.5% statement balance, \$10}	
K	max {3.0% statement balance, \$10}		max {3.0% statement balance, \$10}	

After 2019, lenders typically $\uparrow$ 0.5 pps each year to reach 5% in August 2025

New cards opened from August 2019: all lenders set minimum payments as:
max {5.0% statement balance, \$10}

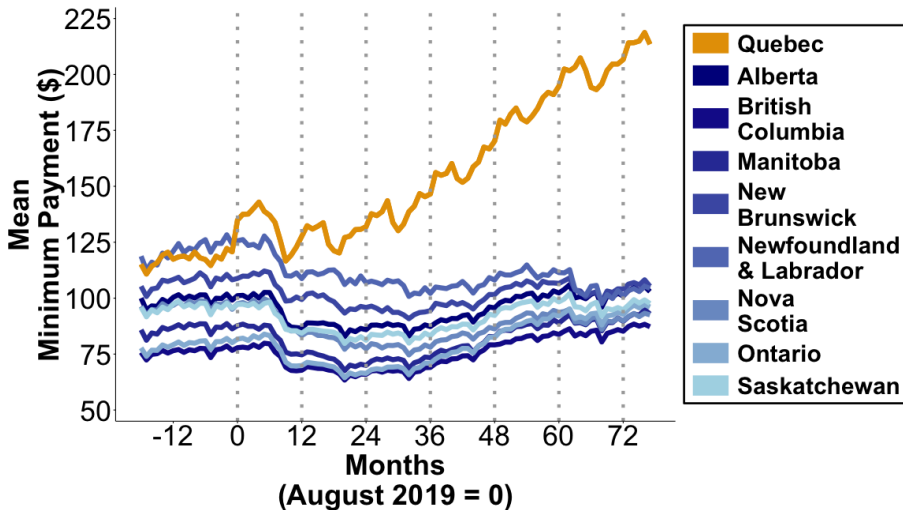
Share of cards with minimum payment:

... $\leq \$10$ 32% in July $\rightarrow$ 19% in August 2019

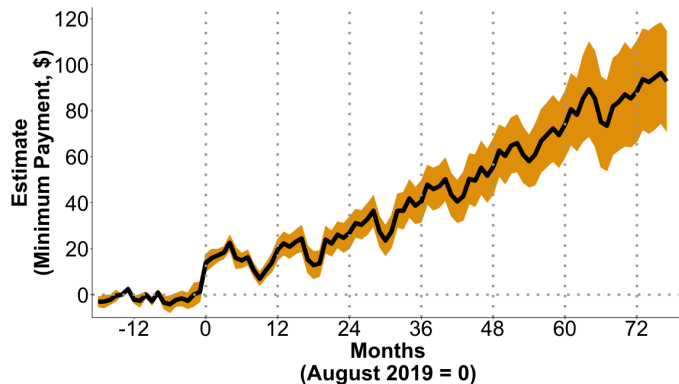
... $< 2\%$: 21% in July $\rightarrow$ $< 2\%$ in August 2019

... $< 5\%$: 62% in July $\rightarrow$ 37% in August 2025

Unconditional Means Show $\uparrow$ Minimum Payments In Quebec, No Increases In Minimum Payments In Other Provinces



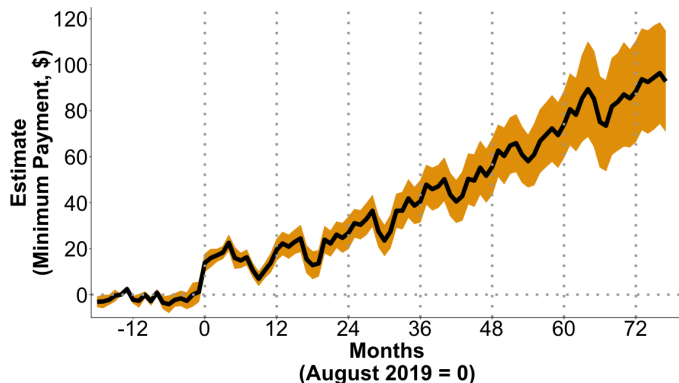
SDID 'First Stage': ↑ Minimum Payments In Quebec 75% After 6 Years



Phase I: ↑ \$14.7 (s.e. \$1.5)

Phase VII: ↑ \$89.3 (s.e. \$11.1)

SDID 'First Stage': $\uparrow$ Minimum Payments In Quebec 75% After 6 Years



Phase I: $\uparrow$ \$14.7 (s.e. \$1.5)

Phase VII: $\uparrow$ \$89.3 (s.e. \$11.1)

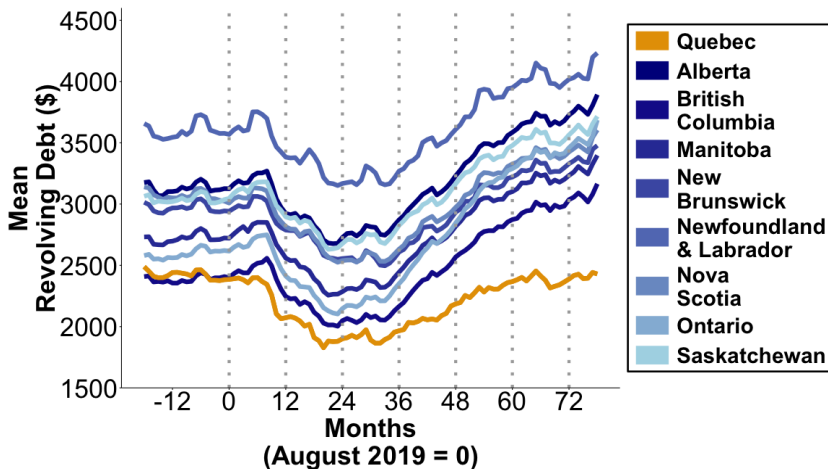
Balance paydown

I: $-\$2.5$ (14% offset)

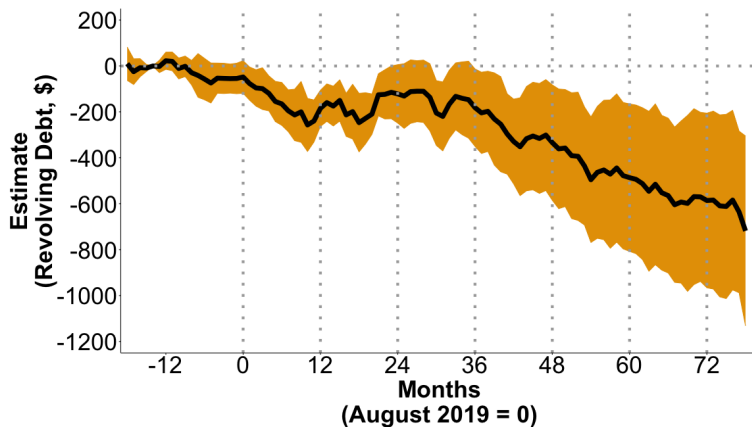
VII: $-\$36$ (29% offset)

Unconditional Means Show ↓ Revolving Debt In Quebec

Revolving Debt (t) = Statement Balance (t-1) - Actual Payments (t)



SDID: ↓ Credit Card Revolving Debt 26% After 6 Years

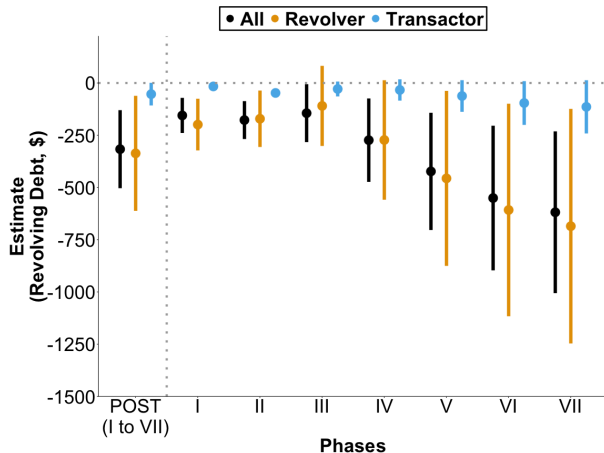


Phase I: ↓ \$155.2 (s.e. \$42.9)

Phase VII: ↓ \$618.5 (s.e. \$197.5)

Baseline mean: \$2,400

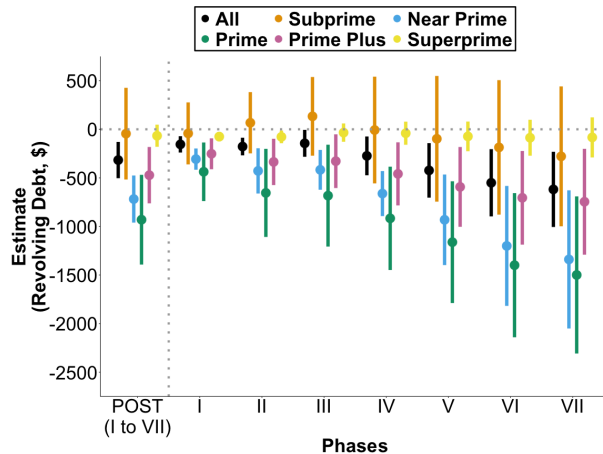
SDID: Overall Effect On Revolving Debt Driven By Pre-Policy Revolvers



Phase I: $-\$199$ is 2x required-paydown ceiling (upper-bound if only paid the minimum)

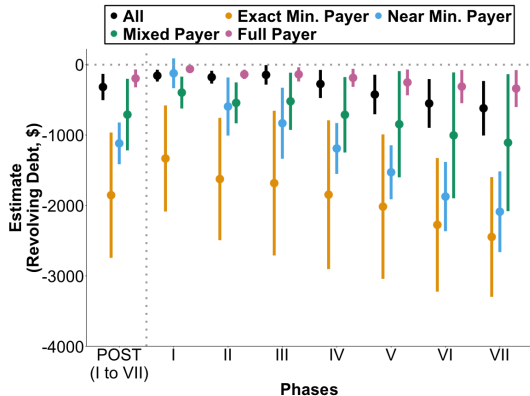
Phase VII: $-\$685.0$ (s.e. $\$286$) $\downarrow$ 23% relative to baseline mean: $\$2,953$

SDID: Overall Effect On Revolving Debt By Pre-Policy Credit Score

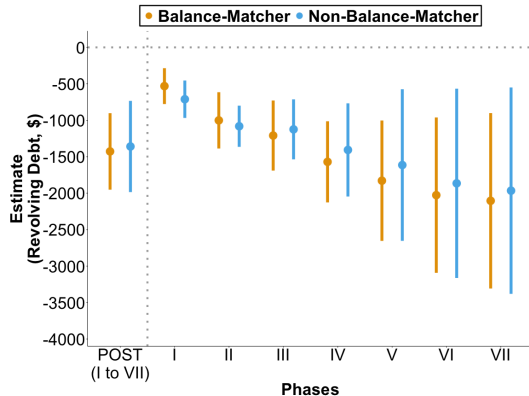


SDID: Unconstrained / Behavioral Consumers Substantially Reduce Debt

By Pre-Policy Payer Type



By Pre-Policy Balance-Matching

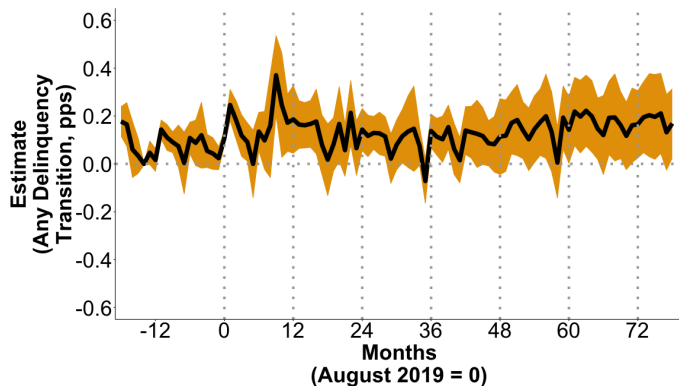


Consistent with minimum payment acting as a reference point

(Keys & Wang, 2019; Medina & Negrin, 2022; Bartels, Herzog, & Sussman, 2024; Guttman-Kenney, 2026; Katz, Russel, & Shi, 2026; Schwartz, 2026)

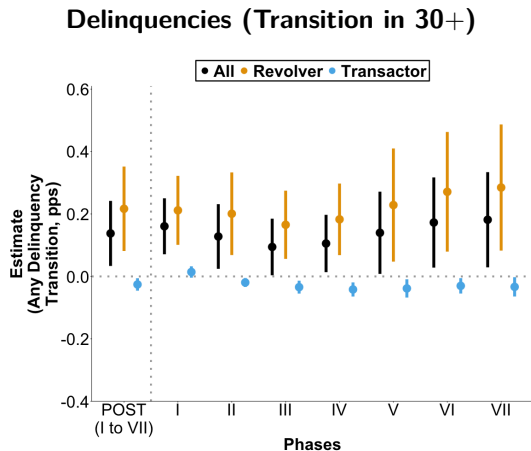
SDID: Any Transition Into Credit Card Delinquency $\uparrow$ 29% After One Month

Delinquency Transition: Any account becomes 30+ Days Past Due

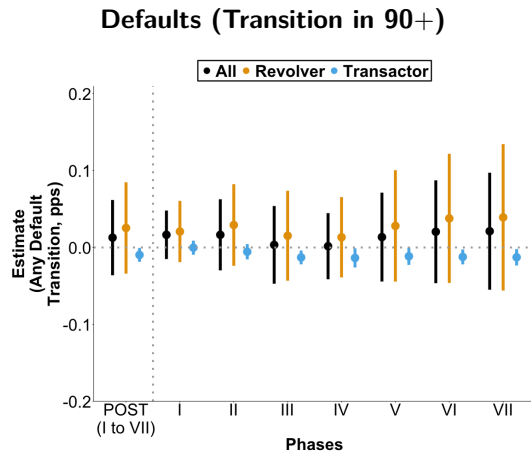


$t+1$: $\uparrow$ 0.25 on a baseline 0.84 percentage points

SDID: No Increases In Credit Card Defaults



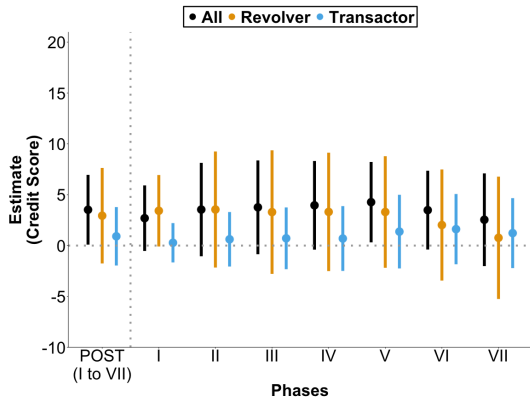
Baseline mean: 0.84%



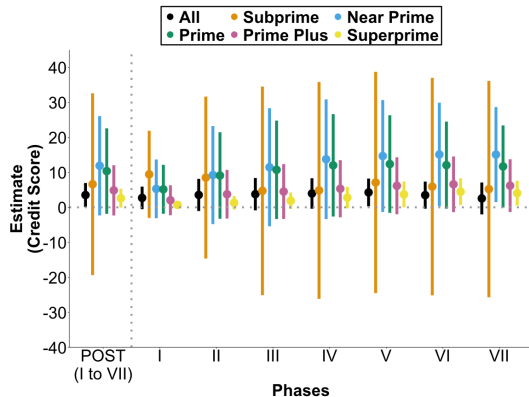
Baseline mean: 0.23%

SDID: Limited Changes In Credit Scores, Slight Improvements

By Pre-Policy Transacting Status

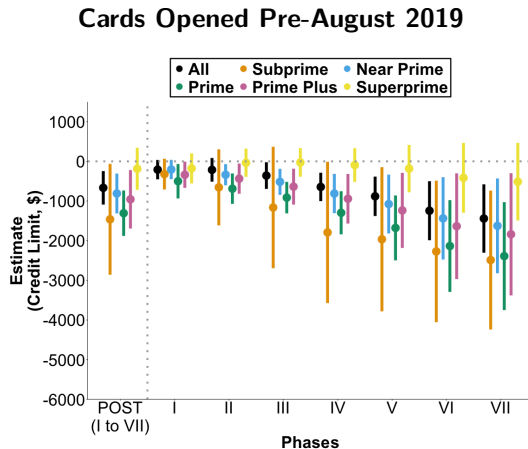
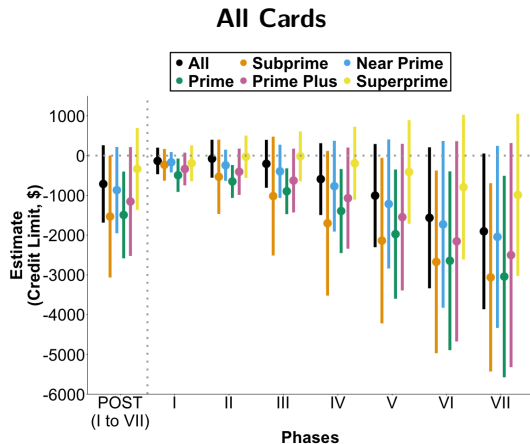


By Pre-Policy Credit Score



Credit Supply

SDID: Some Evidence Of ↓ Credit Card Limits In Long-Run



Baseline mean: \$12,679

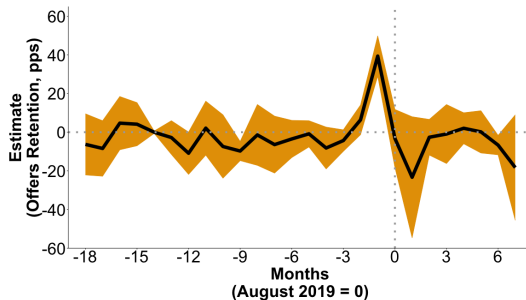
↓ New Credit Cards Opened & ↓ Value of New Credit Card Limits In Quebec

Phase	Number of New Credit Cards Opened Per 100 Consumers (1)	Value (\$) of New Credit Card Limits Per 100 Consumers (2)	Mean Credit Card Limit (\$) of New Cards Opened (\$) (3)	Mean Credit Score of New Cards Opened (4)
I	-0.136 (0.081)	-342 (376)	-51 (139)	0.46 (4.68)
VII	-0.255 (0.082)	-1,589 (591)	-963 (502)	10.50 (5.68)
I to VII	-0.279 (0.052)	-657 (340)	265 (276)	4.14 (4.73)
Baseline Mean	1.779	7,038	3,956	771.66

Consistent with higher minimum payments making some consumers unprofitable

Mintel Data Shows Pre-Policy Spike In Mailed Retention Offers In Quebec

Why? More profitable for lenders to keep consumers on low minimum payment cards opened just before August 2019, than those opened after



Card offers in Quebec just before August 2019 appear to have
↑ credit limits, ↓ APRs, and ↓ annual fees

Theory Model Of Credit Card Issuers

3-period credit supply model. Issuers face heterogeneous fixed cost from changing minimum

Model explains three empirical findings:

1. Reduction in credit limits
2. Some issuers set minimums in Quebec strictly above the regulatory floor
3. Policy increasing delinquency without raising default

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Results

- Credit rationed to revolvers as policy reduces their profitability through (i) lost-interest revenue, and (ii) increased default risk for borrowers near payment capacity
- Served borrowers face a higher per-borrower default probability when minimum payments rise
- Contemporaneous rationing of the riskiest applicants approximately offsets this
- Consistent with composition channel, over-compliers have $\uparrow$ transactors & $\downarrow$ revolving debt

Welfare

Estimated Long-Term Aggregated Impacts In Quebec

Profit Component	Aggregate Impact (Annualized)
Interest Revenue	−\$721 mn [−\$1,172.5 mn, −\$269.8 mn]
Late Fee Revenue	+\$3.8 mn [+\$0.6 mn, +\$7.0 mn]
Charge-offs	≈ \$0
Profit	≈ −\$717.3 mn [−\$1,171.9 mn, −\$262.8 mn]

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How could lenders offset this lost revenue?

- **Higher interest rates?**
 - Lenders already optimize interest rates for credit risk (Edelberg, 2006; Nelson, 2025)
 - Higher rates may ↓ profits by ↓ revolving debt & spending (Bräuning & Stavins, 2026)
- **Higher interchange revenue?** Very unlikely
 - No evidence of increased spending
 - Stricter interchange regulation than in U.S.

Sufficient Statistics Approach To Evaluate Consumer Welfare

Evaluate with behavioral sufficient statistics approach

(Chetty, 2009; Bernheim & Taubinsky, 2018; Allcott et al., 2025; Choukhmane & Palmer, 2026)

Present bias well-documented among credit cardholders

(Meier & Sprenger, 2010; Kuchler & Pagel, 2021; Lee & Maxted, 2026)

- Calculate break-even cost of lost credit access to keep consumer welfare unchanged

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(Meier & Sprenger, 2010; Kuchler & Pagel, 2021; Lee & Maxted, 2026)

- Calculate break-even cost of lost credit access to keep consumer welfare unchanged
- γ is long-run self's dollar loss from last dollar of revolving debt:

$$\gamma = (1 - \beta) \underbrace{\delta(1 + r) \left(\frac{c_2}{c_1} \right)^{-\sigma}}_{\approx 1} \approx 1 - \beta, \quad (16)$$

- Show results across range of external estimates of β
(Allcott et al., 2022; Choukhmane & Palmer, 2026; Lee & Maxted, 2026)

Sufficient Statistics Approach Accounting For Supply Response

- Calculate break-even cost of lost credit access to keep consumer welfare unchanged
 - Pre-policy payer type θ , where $\phi(\theta)$ is size of each type
- **Benefit** is only measured for unconstrained revolving types (near-min. & mixed payers)
- **Access cost** (Λ) includes all types (exact min., near-min., mixed, & full payers)

$$\Delta W = \sum_{\theta} \gamma(-\Delta D^{\text{bias}}(\theta)) \phi(\theta) - \Lambda, \quad \Lambda^{\text{be}} = \sum_{\theta} \gamma(-\Delta D^{\text{bias}}(\theta)) \phi(\theta) \quad (17)$$

- $\Delta D^{\text{bias}}(\theta)$ is the Phase VII change in revolving debt **net of concurrent credit limit cut**
 - Results robust to how calculate limit cut (model-implied or externally-estimated MPBs)

$$\Delta D^{\text{bias}}(\theta) = \Delta D^{\text{obs}}(\theta) - \frac{\partial D}{\partial L} \Delta L(\theta) \quad (22)$$

Break-Even Access Cost

β	$\gamma = 1 - \beta$	Break-even access cost (Λ^{be} , \$ per consumer in Quebec)		
		No supply correction (1)	Supply-corrected, external MPB (2)	Supply-corrected, model-implied (3)
0.55	0.45	148	54	43
0.74	0.26	85	31	25
0.83	0.17	56	20	16

– External MPB: Agarwal et al. (2018) by credit score
(0.59 for subprime/near-prime, 0.28 for prime, 0 for prime-plus/superprime)

– Model-implied: $\partial D / \partial L = \alpha(\theta)(1 - \bar{m}_T) \approx 0.43\text{--}0.58$

- Without supply-correction: \$56 to \$148 per consumer in Quebec: \$327 to \$863 million
- With supply-correction: \$16 to \$54 per consumer in Quebec: \$93 to \$315 million

Conclusions

Conclusions

- Use comprehensive Canadian credit reporting data and a synthetic difference-in-differences design to estimate the causal effects of a hard paternalistic policy
- Quebec policy permanently ↑ credit card minimum payments by 75% after 6 years
 - Permanently ↓ credit card debt by 26% after 6 years
 - Driven by pre-policy revolvers
 - Large responses by unconstrained / behavioral consumers
 - Temporarily ↑ delinquency by 29% for one month
 - No increase in defaults
 - Permanently ↓ credit card access
- Strategic marketing by lenders to keep consumers on old cards with lower minimums
- Consumer welfare rises unless lost access worth over \$16 to \$54 per Quebec consumer (compared to over \$56 to \$148 without supply correction)

Thank you!



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Quebec Minimum Payment Policy (“Bill 134”)

Minimum Payment

Every month, you have to reimburse the amount that appears on your credit card statement, or a part of that amount.

If you only reimburse part of the balance, you have to pay the “minimum payment” or “minimum instalment.” The way to determine this amount is indicated in the contract, generally expressed as a percentage of the balance owing.

Amount of the minimum payment

The *Consumer Protection Act* provides for the minimum payment percentage you can be charged.

Contract entered into on or after August 1, 2019

If you entered into a credit card contract on or after August 1, 2019, the minimum payment to be made every month must correspond to **at least 5%** of the balance owing indicated on your account statement.

Contract entered into before August 1, 2019

If you already had a credit card before August 1, 2019 and the minimum payment percentage was set at less than 2%, **for the period from August 1, 2019 to July 31, 2020**, the minimum payment to be made every month must correspond to **2%** of the balance owing indicated on your account statement.

What if the issuer of your credit card claims more than 2%?

- They may do so if the contract already provided for that possibility.
- They may not do so if they amended the contract so as to charge more than 2% without your consent.

Over the coming years, the minimum payment percentage will increase. Credit card issuers will have to charge a minimum payment that corresponds to at least the following percentage of the balance owing:

- 2.5%, as of August 1, 2020;
- 3%, as of August 1, 2021;
- 3.5%, as of August 1, 2022;
- 4%, as of August 1, 2023;
- 4.5%, as of August 1, 2024;
- 5%, as of August 1, 2025.

If the credit card issuer charges you a percentage that is higher than what is provided for in the contract or by the Act, [contact the Office de la protection du consommateur](#).

Weights for Each Province in SDID 🗺️

Province	Province Weight ($\hat{\omega}_p$)
Quebec	1.0000
Alberta	0.1252
British Columbia	0.1185
Manitoba	0.1131
New Brunswick	0.1353
Newfoundland & Labrador	0.1364
Nova Scotia	0.1372
Ontario	0.1225
Saskatchewan	0.1118

Shares of Consumers in Heterogeneous Groups

Heterogeneous group	Share (%)
Transacting Status	
Revolver	77.78
Transactor	22.22
Credit Score	
Subprime	8.68
Near Prime	12.68
Prime	12.68
Prime Plus	22.75
Superprime	43.21
Payer Type	
Exact Min. Payer	1.00
Near Min. Payer	3.86
Mixed Payer	22.41
Full Payer	72.72
Balance-Matching	
Balance-Matcher	18.28
Non-Balance-Matcher	81.72

Lender Minimum Payment Formulas

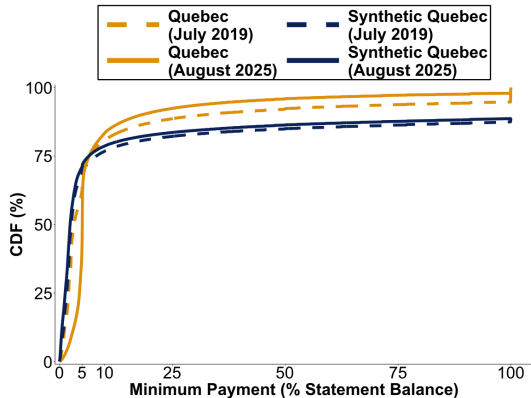
Issuer	Quebec Cards in July 2019 & Cards in Rest of Canada from July 2019 onward	Quebec Cards Opened Before August 2019						
		Phase I August 2019	Phase II August 2020	Phase III August 2021	Phase IV August 2022	Phase V August 2023	Phase VI August 2024	Phase VII August 2025
A	\$10 + interest + fees	max{2.0% B, \$10}	max{2.5% B, \$10}	max{3.0% B, \$10}	max{3.5% B, \$10}	max{4.0% B, \$10}	max{4.5% B, \$10}	max{5.0% B, \$10}
B	\$10 + interest + fees	max{2.0% B, \$10}	max{2.5% B, \$10}	max{3.0% B, \$10}	max{3.5% B, \$10}	max{4.0% B, \$10}	max{4.5% B, \$10}	max{5.0% B, \$10}
C	\$10 + interest + fees	max{2.5% B, \$10}	max{2.5% B, \$10}	max{3.0% B, \$10}	max{3.5% B, \$10}	max{4.0% B, \$10}	max{4.5% B, \$10}	max{5.0% B, \$10}
D	\$10 + interest + fees	max{2.5% B, \$10}	max{2.5% B, \$10}	max{3.0% B, \$10}	max{3.5% B, \$10}	max{4.0% B, \$10}	max{4.5% B, \$10}	max{5.0% B, \$10}
E	\$10 + interest + fees	max{2.5% B, \$10}	max{3.0% B, \$10}	max{3.5% B, \$10}	max{4.0% B, \$10}	max{4.5% B, \$10}	max{5.0% B, \$10}	max{5.0% B, \$10}
F	\$10 + interest + fees	max{3.0% B, \$10}	max{3.0% B, \$10}	max{3.0% B, \$10}	max{3.5% B, \$10}	max{4.0% B, \$10}	max{4.5% B, \$10}	max{5.0% B, \$10}
G	max{1.0% B, \$10}	max{3.0% B, \$10}	max{3.0% B, \$10}	max{3.0% B, \$10}	max{3.5% B, \$10}	max{4.0% B, \$10}	max{4.5% B, \$10}	max{5.0% B, \$10}
H	max{2.0% B, \$10}	max{2.0% B, \$10}	max{2.5% B, \$10}	max{3.0% B, \$10}	max{3.5% B, \$10}	max{4.0% B, \$10}	max{4.5% B, \$10}	max{5.0% B, \$10}
I	max{2.0% B, \$10}	max{2.0% B, \$10}	max{2.5% B, \$10}	max{3.0% B, \$10}	max{3.5% B, \$10}	max{4.0% B, \$10}	max{4.5% B, \$10}	max{5.0% B, \$10}
J	max{2.5% B, \$10}	max{2.5% B, \$10}	max{2.5% B, \$10}	max{3.0% B, \$10}	max{3.5% B, \$10}	max{4.0% B, \$10}	max{4.5% B, \$10}	max{5.0% B, \$10}
K	max{3.0% B, \$10}	max{3.0% B, \$10}	max{3.0% B, \$10}	max{3.0% B, \$10}	max{3.5% B, \$10}	max{4.0% B, \$10}	max{4.5% B, \$10}	max{5.0% B, \$10}

“B” is outstanding balance (statement balance inclusive of interest and fees)

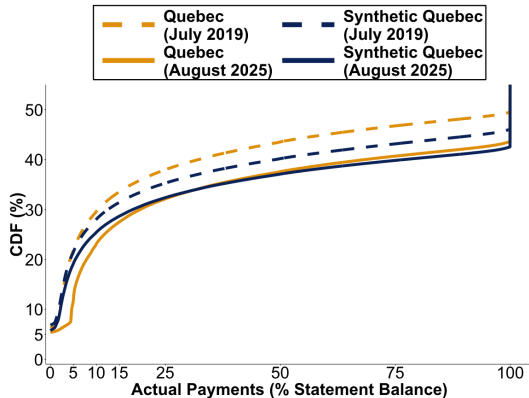
- All lenders set minimum payments on new cards opened in Quebec from August 2019:
max {5.0% statement balance, \$10}
- This formula produces the lowest minimum payment to comply with the policy

Distribution of minimum payments and actual payments 🚀

Minimum Payments (% Balance)

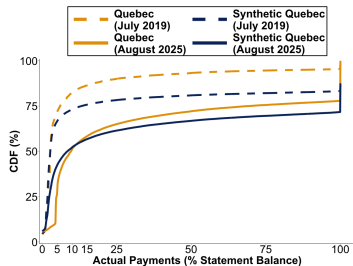


Actual Payments (% Balance)

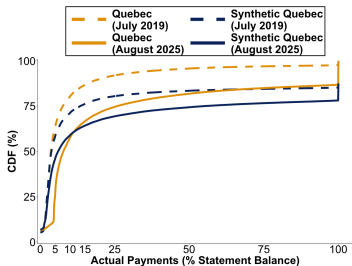


Distribution of actual payments for each pre-policy payer type 🚀

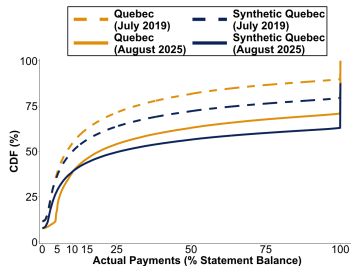
A. Exact minimum payers



B. Near minimum payers



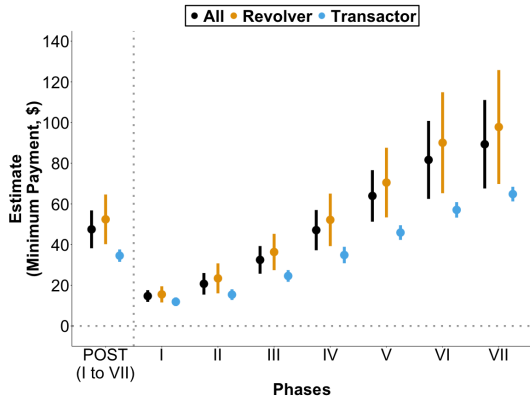
C. Mixed payers



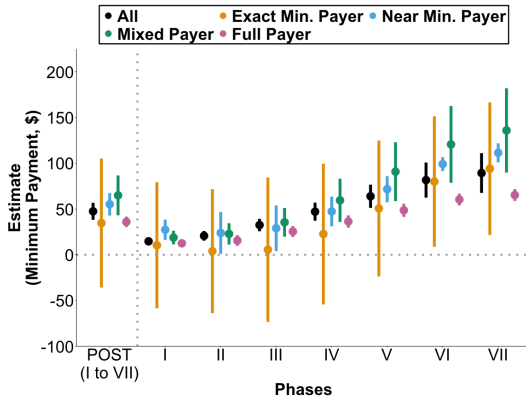
SDID: Heterogeneous Effects on Minimum Payments



By Pre-Policy Transacting Status



By Pre-Policy Payer Type



Methodology for Decomposing the Effect on Minimum Payments

Construct, for each policy phase j , a counterfactual minimum payment ($\hat{m}_{i,t}$) each card-month:

$$\hat{m}_{i,t} = \max \{ f_l^j b_{i,t}, \$10 \}. \quad (18)$$

Rate f_l^j is the actual minimum payment formula of issuer l in Quebec during phase j

Aggregate to estimate SDID outcome $\delta_j^{PHASE}(\hat{m}_{p,t})$

– **Balance channel** captures debt paydown (formula held fixed & only balances move)

Let $g = \hat{m}_{p,t} - m_{p,t}$ be the gap between the constructed and observed minimum

$$\delta_j^{PHASE}(m_{p,t}) = \delta_j^{PHASE}(\hat{m}_{p,t}) + \underbrace{[\bar{y}_j^{SQC}(g) - \bar{y}_{\text{baseline}}^{SQC}(g)]}_{\tilde{M}_j(\text{mechanical})} + \underbrace{[\bar{y}_{\text{baseline}}^{QC}(g) - \bar{y}_j^{QC}(g)]}_{\tilde{A}_j(\text{Quebec wedge})}, \quad (19)$$

Mechanical term, $\tilde{M}_j$, is the growth of the formula-versus-observed gap in synthetic Quebec

Wedge, $\tilde{A}_j$, is the residual part of Quebec's increase (e.g., level differences in balances)

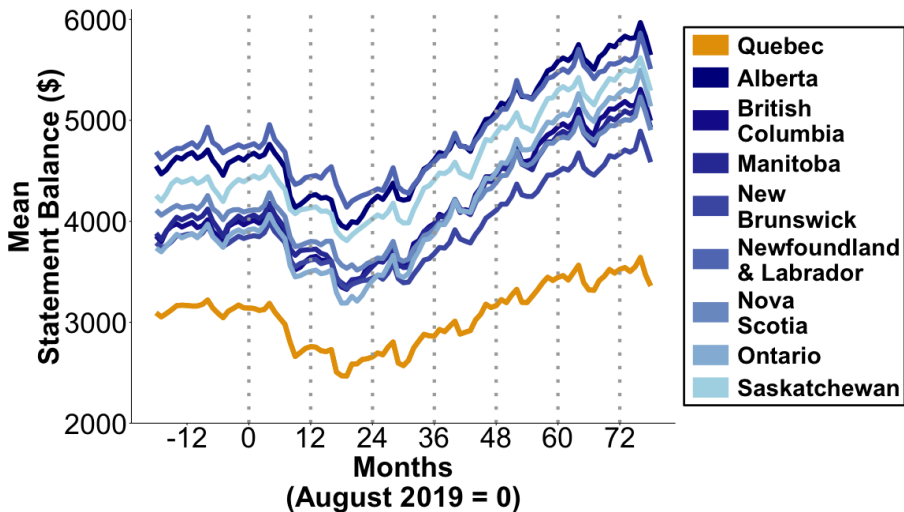
Decompose the effects of Bill 134 on minimum payments into:

- Part driven by changes in the issuer's formula, which increases minimum payments
- **Balance Channel:** the part driven by consumers accelerating the pay down of their balances, which then lowers subsequent minimum payments

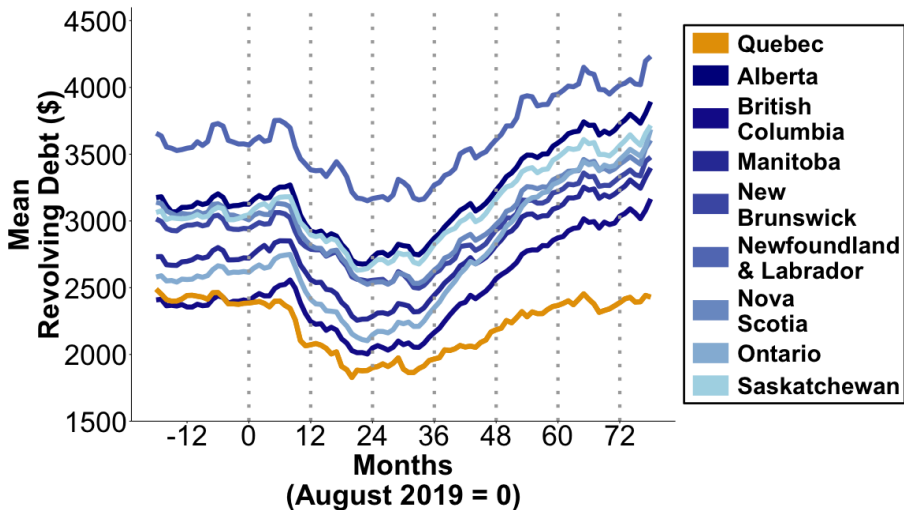
Change in mean minimum payments due to balance channel:

- Phase I: −\$2.5 per month (14% offset)
- Phase VII: −\$36 per month (29% offset)

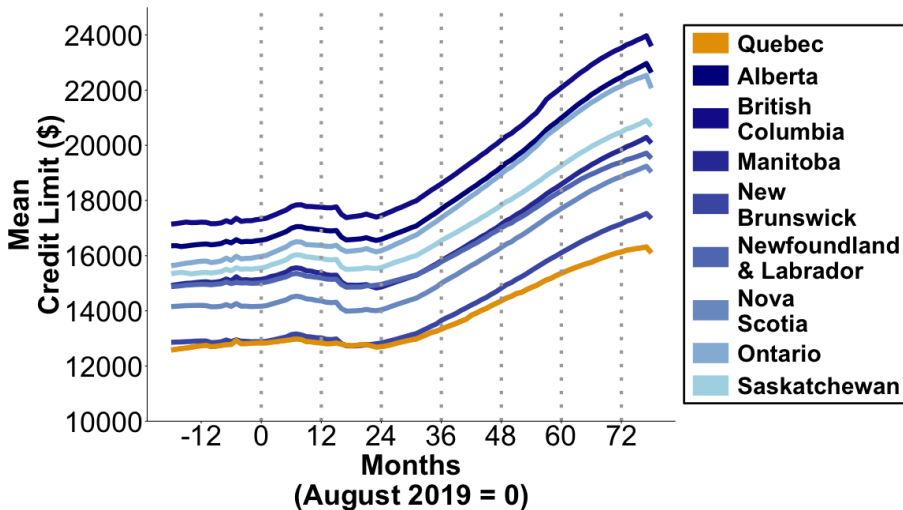
Unconditional Means: Statement Balance 📍



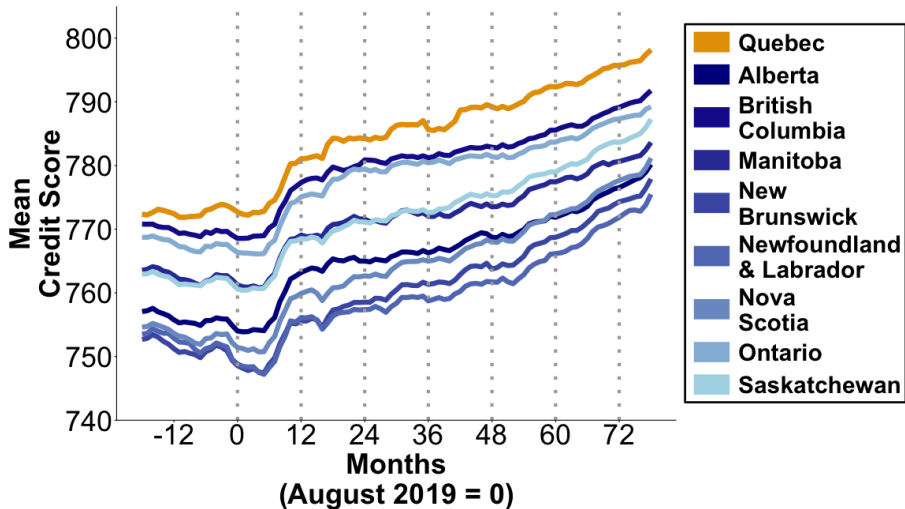
Unconditional Means: Revolving Debt 📍



Unconditional Means: Credit Limit 📍



Unconditional Means: Credit Score 📍

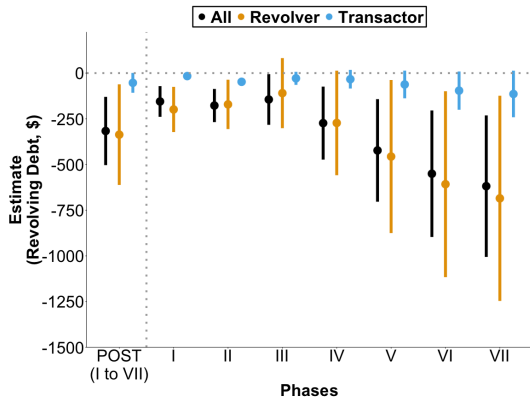


Required-Paydown Ceiling: Methodology

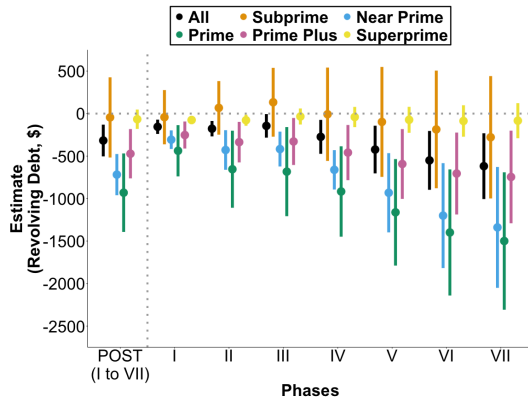
- Ceiling: cumulative forced minimum payments, holding spending at its counterfactual path
- Event month τ : $\text{Ceiling}_{\tau} = \delta_I^{\min} \times (\tau + 1)$
- Phase I, revolvers: $\delta_I^{\min} = \$15.53/\text{month}$
 - Month 0: \$15.53
 - Month 11: $\$15.53 \times 12 = \186.36
- δ^{PHASE} averages over 12 months:
$$\$15.53 \times \frac{1 + 2 + \dots + 12}{12} = \$15.53 \times 6.5 = \$101$$
- Estimated revolving debt effect ($-\$199$) is $\approx 2 \times$ this ceiling
- Spending cannot overturn the ceiling:
 - Overspending only shrinks the measured decline
 - Underspending is a voluntary response

SDID: Heterogeneous Effects on Revolving Debt 🚀

By Pre-Policy Transacting Status

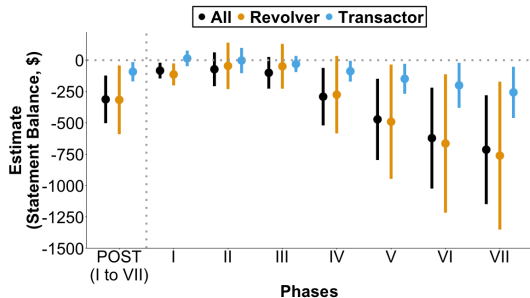


By Pre-Policy Credit Score

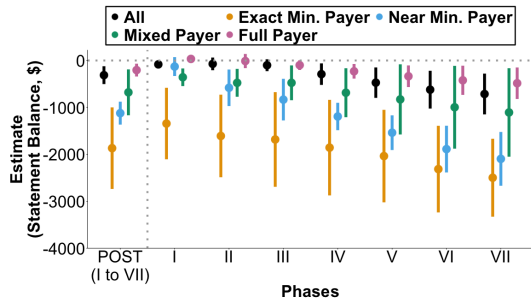


SDID: Heterogeneous Effects on Statement Balance 🚀

By Pre-Policy Transacting Status

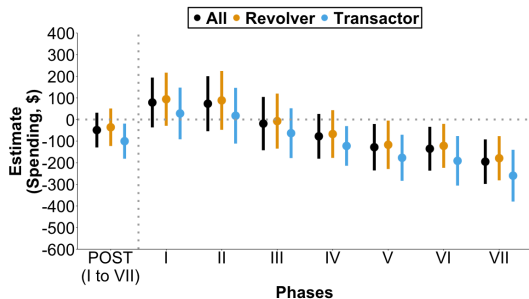


By Pre-Policy Payer Type

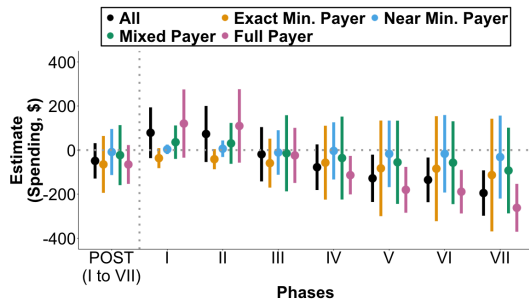


SDID: Heterogeneous Effects on Spending 🚀

By Pre-Policy Transacting Status

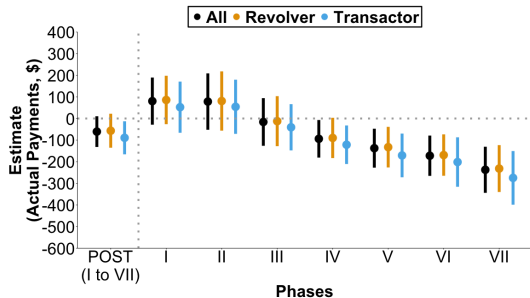


By Pre-Policy Payer Type

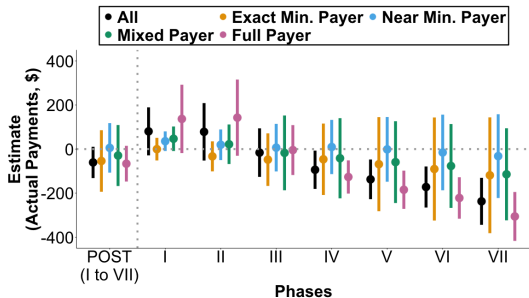


SDID: Heterogeneous Effects on Actual Payments 🚀

By Pre-Policy Transacting Status

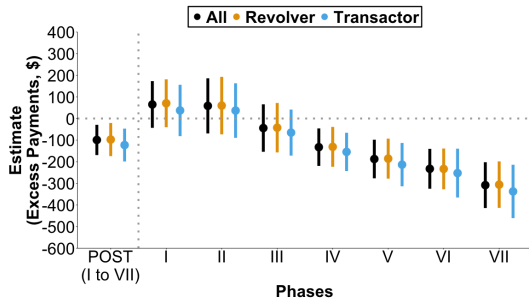


By Pre-Policy Payer Type

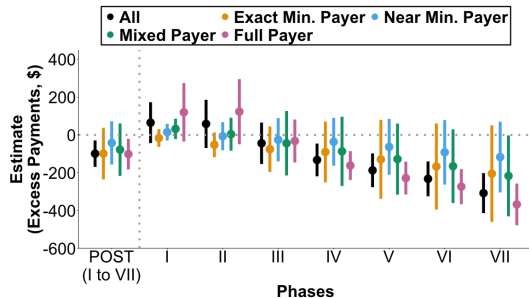


SDID: Heterogeneous Effects on Excess Payments 🚀

By Pre-Policy Transacting Status

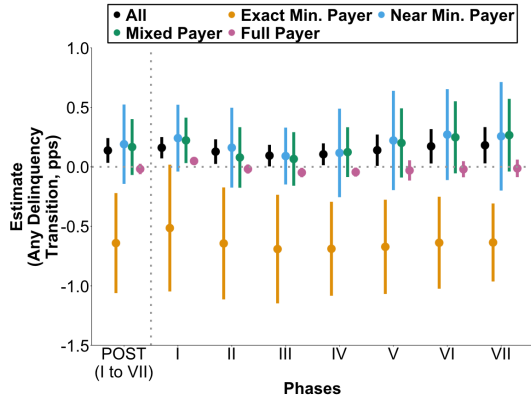


By Pre-Policy Payer Type

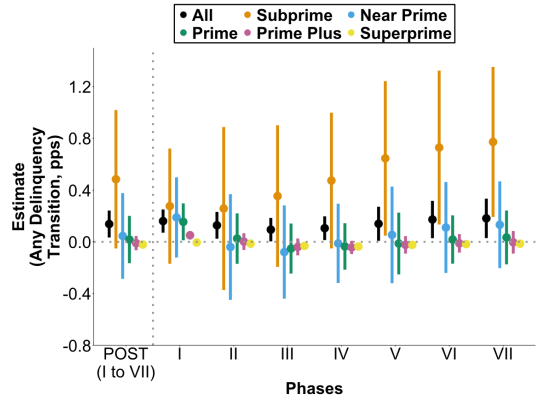


SDID: Heterogeneous Effects on Transitions into Delinquency 🚀

By Pre-Policy Payer Type

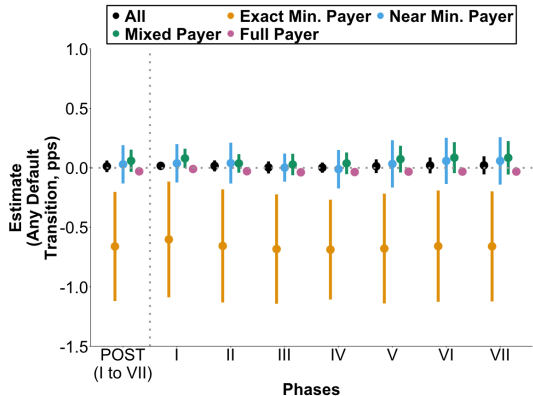


By Pre-Policy Credit Score

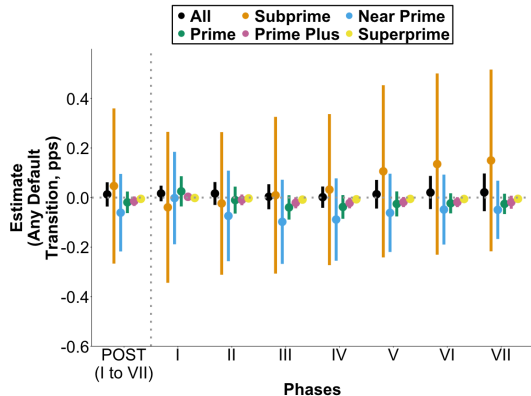


SDID: Heterogeneous Effects on Transitions into Default 🚀

By Pre-Policy Payer Type

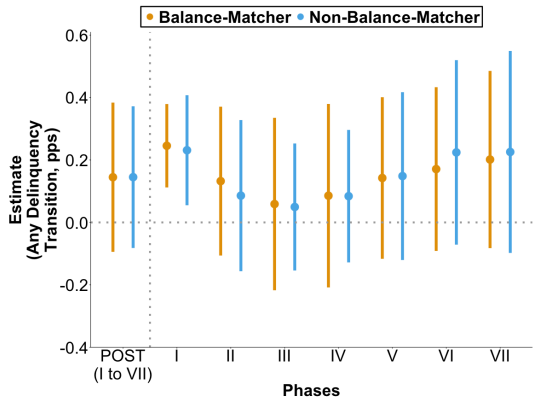


By Pre-Policy Credit Score

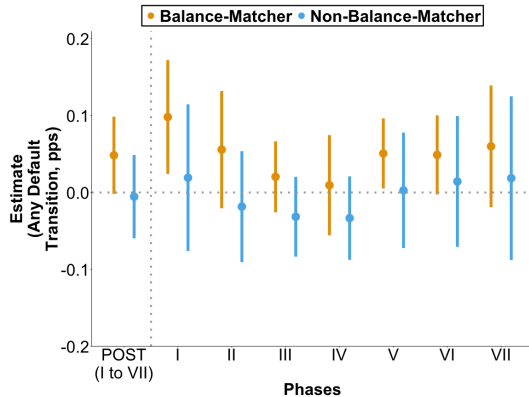


SDID: Heterogeneous Effects by Balance Matching

Transitions into Delinquency

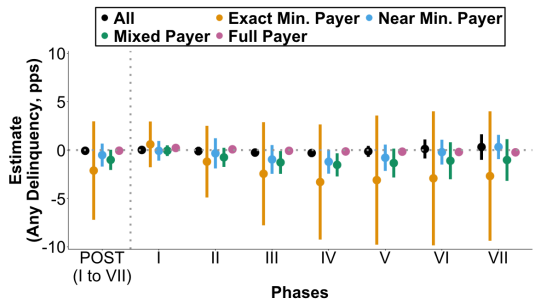


Transitions into Default

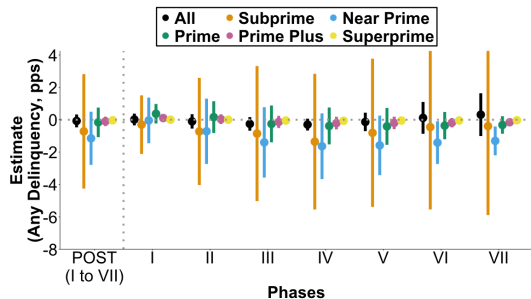


SDID: Heterogeneous Effects on Any Delinquency 🚀

By Pre-Policy Payer Type

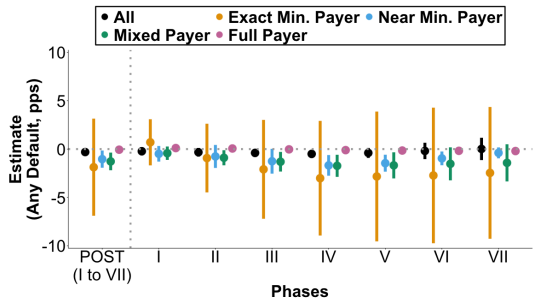


By Pre-Policy Credit Score

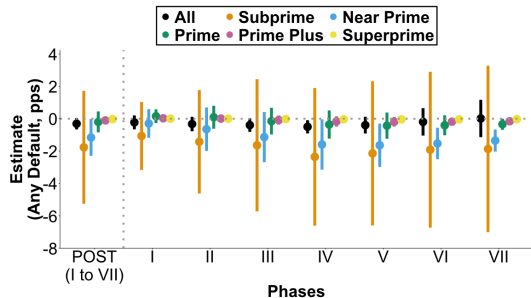


SDID: Heterogeneous Effects on Any Default

By Pre-Policy Payer Type

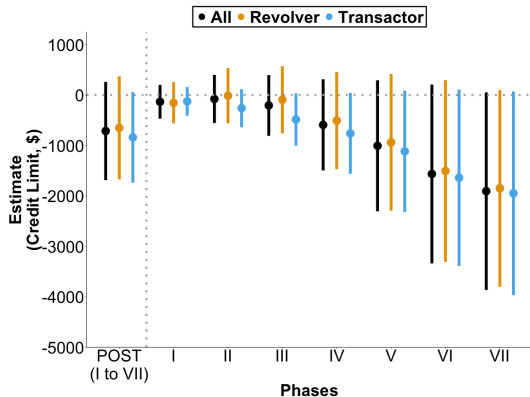


By Pre-Policy Credit Score

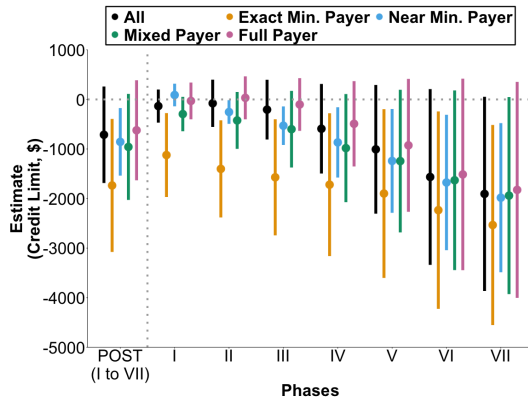


SDID: Heterogeneous Effects on Credit Limits

By Pre-Policy Transacting Status



By Pre-Policy Payer Type



SDID Estimates by Policy Phase

Phase	Minimum Payment (\$) (1)	Revolving Debt (\$) (2)	Statement Balance (\$) (3)	Any Delinquency Transition (pps) (4)	Any Default Transition (pps) (5)	Credit Limit (\$) (6)
I	14.72 (1.47)	-155.15 (42.86)	-82.60 (32.14)	0.1606 (0.0457)	0.0165 (0.0161)	-134.00 (170.40)
II	20.72 (2.71)	-177.37 (46.27)	-72.16 (68.80)	0.1279 (0.0528)	0.0165 (0.0236)	-78.01 (243.28)
III	32.48 (3.47)	-144.31 (70.68)	-100.40 (64.48)	0.0944 (0.0462)	0.0034 (0.0258)	-205.81 (307.45)
IV	47.11 (5.05)	-273.53 (101.84)	-290.75 (117.03)	0.1055 (0.0468)	0.0017 (0.0219)	-590.94 (460.90)
V	63.93 (6.46)	-422.99 (143.11)	-472.05 (165.41)	0.1398 (0.0672)	0.0135 (0.0295)	-1006.13 (662.27)
VI	81.63 (9.78)	-550.40 (176.56)	-621.52 (205.43)	0.1727 (0.0737)	0.0205 (0.0341)	-1564.09 (904.54)
VII	89.33 (11.10)	-618.48 (197.49)	-713.70 (221.68)	0.1815 (0.0778)	0.0212 (0.0388)	-1905.31 (999.29)
I to VII	47.50 (4.75)	-316.64 (95.26)	-312.28 (96.97)	0.1377 (0.0531)	0.0128 (0.0250)	-712.47 (496.55)
Baseline Mean	119.31	2399.68	3164.58	0.8407	0.2276	12678.56

SDID Estimates by Policy Phase: Revolvers

Phase	Minimum Payment (\$) (1)	Revolving Debt (\$) (2)	Statement Balance (\$) (3)	Any Delinquency Transition (pps) (4)	Any Default Transition (pps) (5)	Credit Limit (\$) (6)
I	15.53 (2.03)	-198.89 (63.08)	-113.29 (44.79)	0.2117 (0.0564)	0.0207 (0.0203)	-153.28 (208.93)
II	23.40 (3.76)	-171.09 (68.87)	-45.28 (94.54)	0.2008 (0.0675)	0.0292 (0.0271)	-13.02 (280.32)
III	36.36 (4.55)	-109.56 (97.78)	-48.50 (91.23)	0.1654 (0.0557)	0.0153 (0.0298)	-92.95 (338.98)
IV	52.16 (6.59)	-272.59 (145.88)	-274.94 (157.71)	0.1828 (0.0585)	0.0133 (0.0266)	-506.07 (491.09)
V	70.48 (8.74)	-456.36 (213.61)	-490.55 (232.67)	0.2286 (0.0924)	0.0281 (0.0369)	-937.93 (691.46)
VI	90.07 (12.66)	-607.63 (259.43)	-664.42 (281.70)	0.2711 (0.0978)	0.0378 (0.0428)	-1504.92 (918.47)
VII	97.79 (14.29)	-685.03 (286.34)	-761.16 (301.12)	0.2848 (0.1031)	0.0392 (0.0486)	-1848.98 (995.34)
I to VII	52.41 (6.23)	-336.57 (140.41)	-316.10 (139.91)	0.2167 (0.0690)	0.0254 (0.0303)	-651.15 (521.30)
Baseline Mean	138.78	2953.22	3683.18	1.0411	0.2814	12432.02

SDID Estimates by Policy Phase: Transactors

Phase	Minimum Payment (\$) (1)	Revolving Debt (\$) (2)	Statement Balance (\$) (3)	Any Delinquency Transition (pps) (4)	Any Default Transition (pps) (5)	Credit Limit (\$) (6)
I	11.90 (1.09)	-16.27 (10.85)	14.42 (31.85)	0.0142 (0.0093)	-0.0003 (0.0047)	-126.47 (144.92)
II	15.41 (1.34)	-47.60 (10.61)	-2.46 (51.50)	-0.0194 (0.0073)	-0.0056 (0.0051)	-261.05 (193.15)
III	24.59 (1.47)	-28.59 (18.19)	-29.85 (32.78)	-0.0343 (0.0105)	-0.0129 (0.0046)	-485.39 (264.34)
IV	34.87 (2.06)	-33.34 (26.00)	-87.68 (42.34)	-0.0418 (0.0115)	-0.0136 (0.0064)	-759.14 (408.62)
V	45.91 (1.84)	-62.17 (38.62)	-148.22 (61.10)	-0.0384 (0.0149)	-0.0115 (0.0057)	-1115.83 (612.66)
VI	57.07 (1.94)	-95.78 (53.40)	-200.30 (91.63)	-0.0302 (0.0126)	-0.0123 (0.0049)	-1638.29 (891.88)
VII	64.82 (1.84)	-114.09 (64.96)	-256.47 (104.30)	-0.0334 (0.0157)	-0.0127 (0.0054)	-1947.36 (1030.12)
I to VII	34.56 (1.55)	-53.21 (27.67)	-91.70 (39.35)	-0.0257 (0.0102)	-0.0096 (0.0046)	-838.8 (459.11)
Baseline Mean	49.99	428.42	1317.75	0.1270	0.0360	13556.55

SDID Estimates by Policy Phase: Exact Minimum Payers 📍

Phase	Minimum Payment (\$) (1)	Revolving Debt (\$) (2)	Statement Balance (\$) (3)	Any Delinquency Transition (pps) (4)	Any Default Transition (pps) (5)	Credit Limit (\$) (6)
I	10.43 (35.13)	-1331.76 (384.53)	-1343.16 (388.64)	-0.5146 (0.2718)	-0.6018 (0.2482)	-1121.92 (431.72)
II	3.97 (34.56)	-1624.08 (443.26)	-1607.95 (448.75)	-0.6436 (0.2401)	-0.6563 (0.2425)	-1400.79 (499.88)
III	5.57 (40.25)	-1681.74 (524.54)	-1682.63 (514.44)	-0.6911 (0.2329)	-0.6825 (0.2345)	-1570.96 (597.33)
IV	22.67 (39.26)	-1846.30 (538.95)	-1856.60 (519.37)	-0.6885 (0.2015)	-0.6872 (0.2140)	-1719.29 (734.98)
V	50.64 (37.9)	-2015.76 (523.35)	-2036.69 (502.83)	-0.6724 (0.2021)	-0.6785 (0.2354)	-1898.18 (868.38)
VI	80.00 (36.39)	-2274.45 (484.14)	-2314.10 (470.82)	-0.6378 (0.1972)	-0.658 (0.2385)	-2232.97 (1015.80)
VII	94.15 (36.96)	-2446.52 (433.80)	-2497.98 (423.53)	-0.6357 (0.1672)	-0.6601 (0.2359)	-2533.29 (1029.73)
I to VII	34.66 (35.95)	-1853.35 (454.16)	-1868.10 (443.30)	-0.6408 (0.2144)	-0.6607 (0.2340)	-1734.96 (684.42)
Baseline Mean	265.26	8270.97	8542.24	2.3188	2.1899	12719.54

SDID Estimates by Policy Phase: Near Minimum Payers

Phase	Minimum Payment (\$) (1)	Revolving Debt (\$) (2)	Statement Balance (\$) (3)	Any Delinquency Transition (pps) (4)	Any Default Transition (pps) (5)	Credit Limit (\$) (6)
I	27.34 (5.66)	-121.82 (108.11)	-129.25 (102.62)	0.2413 (0.1432)	0.0382 (0.0823)	89.48 (116.43)
II	23.87 (11.69)	-593.94 (210.49)	-581.36 (198.61)	0.1612 (0.1712)	0.0397 (0.0876)	-253.52 (121.67)
III	29.00 (12.74)	-831.12 (257.21)	-833.70 (226.26)	0.0908 (0.1219)	0.0020 (0.0603)	-529.60 (198.28)
IV	47.35 (8.27)	-1189.54 (184.34)	-1193.07 (148.93)	0.1170 (0.1898)	-0.0112 (0.0825)	-866.60 (361.74)
V	71.66 (7.26)	-1527.93 (195.64)	-1538.04 (188.50)	0.2217 (0.2129)	0.0331 (0.1013)	-1240.02 (534.14)
VI	99.11 (3.87)	-1873.07 (251.17)	-1887.88 (253.95)	0.2710 (0.1949)	0.0582 (0.0991)	-1675.23 (696.70)
VII	111.38 (5.33)	-2088.59 (292.46)	-2095.82 (292.33)	0.2568 (0.2327)	0.0584 (0.1017)	-1981.85 (767.42)
I to VII	55.18 (6.27)	-1117.33 (151.26)	-1121.9 (124.6)	0.1903 (0.1701)	0.0295 (0.0822)	-855.43 (347.59)
Baseline Mean	196.40	6521.15	6874.71	1.6671	0.3397	10663.01

SDID Estimates by Policy Phase: Mixed Payers 🚀

Phase	Minimum Payment (\$) (1)	Revolving Debt (\$) (2)	Statement Balance (\$) (3)	Any Delinquency Transition (pps) (4)	Any Default Transition (pps) (5)	Credit Limit (\$) (6)
I	18.84 (3.79)	-395.94 (115.58)	-357.11 (95.80)	0.2225 (0.0974)	0.0796 (0.0413)	-294.96 (176.94)
II	22.83 (5.93)	-542.21 (148.22)	-475.45 (151.47)	0.0789 (0.1298)	0.0370 (0.0398)	-423.13 (292.70)
III	35.50 (7.99)	-519.83 (207.38)	-475.49 (189.04)	0.0664 (0.1149)	0.0279 (0.0460)	-600.28 (394.27)
IV	59.50 (12.05)	-711.53 (273.27)	-686.56 (266.17)	0.1236 (0.1068)	0.0380 (0.0465)	-981.88 (556.41)
V	90.84 (16.40)	-845.45 (384.86)	-828.72 (381.36)	0.2010 (0.1484)	0.0733 (0.0573)	-1243.56 (733.96)
VI	120.62 (21.40)	-1003.39 (455.98)	-996.37 (450.08)	0.2482 (0.1545)	0.0857 (0.0664)	-1630.13 (924.45)
VII	135.89 (23.48)	-1107.23 (496.48)	-1106.99 (481.01)	0.2661 (0.1558)	0.0846 (0.0718)	-1939.57 (1013.87)
I to VII	64.92 (11.10)	-708.49 (259.41)	-678.3 (249.39)	0.1664 (0.1198)	0.0594 (0.0477)	-957.77 (545.52)
Baseline Mean	200.76	5530.06	6350.49	2.1580	0.5391	12114.82

SDID Estimates by Policy Phase: Full Payers

Phase	Minimum Payment (\$) (1)	Revolving Debt (\$) (2)	Statement Balance (\$) (3)	Any Delinquency Transition (pps) (4)	Any Default Transition (pps) (5)	Credit Limit (\$) (6)
I	12.54 (1.70)	-60.16 (24.59)	33.73 (36.61)	0.0490 (0.0164)	-0.0098 (0.0042)	-29.91 (189.10)
II	15.54 (3.00)	-136.57 (33.56)	-11.70 (77.04)	-0.0182 (0.0196)	-0.0285 (0.0059)	32.26 (221.05)
III	25.45 (3.02)	-137.48 (51.36)	-98.80 (52.71)	-0.0478 (0.0203)	-0.0375 (0.0072)	-102.07 (271.21)
IV	36.35 (3.43)	-186.31 (65.34)	-229.92 (79.48)	-0.0442 (0.0193)	-0.0355 (0.0067)	-491.49 (439.00)
V	48.65 (3.67)	-251.28 (93.40)	-335.51 (117.60)	-0.0300 (0.0434)	-0.0322 (0.0078)	-925.86 (683.86)
VI	60.38 (3.27)	-310.85 (120.58)	-421.68 (158.63)	-0.0201 (0.0345)	-0.0316 (0.0091)	-1512.62 (985.40)
VII	65.31 (3.31)	-338.69 (133.93)	-484.24 (171.27)	-0.0129 (0.0376)	-0.0314 (0.0105)	-1824.27 (1111.38)
I to VII	36.00 (2.90)	-194.46 (64.67)	-204.51 (69.63)	-0.0180 (0.0223)	-0.0294 (0.0062)	-621.85 (514.77)
Baseline Mean	86.85	1086.17	1862.98	0.3522	0.0912	12965.71

SDID Estimates by Policy Phase: Balance-Matchers

Phase	Minimum Payment (\$) (1)	Revolving Debt (\$) (2)	Statement Balance (\$) (3)	Any Delinquency Transition (pps) (4)	Any Default Transition (pps) (5)	Credit Limit (\$) (6)
I	39.64 (10.11)	-531.30 (125.40)	-471.07 (159.15)	0.2456 (0.0681)	0.0981 (0.0377)	-331.34 (257.39)
II	40.89 (19.21)	-1000.97 (196.91)	-840.13 (246.97)	0.1324 (0.1216)	0.0557 (0.0389)	-416.28 (383.48)
III	52.55 (17.25)	-1209.15 (245.20)	-1056.87 (223.56)	0.0589 (0.1409)	0.0204 (0.0235)	-510.41 (461.96)
IV	79.14 (10.75)	-1569.95 (283.89)	-1418.54 (285.52)	0.0856 (0.1499)	0.0094 (0.0332)	-775.72 (660.99)
V	114.23 (9.95)	-1828.43 (421.16)	-1669.44 (432.72)	0.1423 (0.1320)	0.0507 (0.0233)	-1072.99 (884.85)
VI	152.27 (12.58)	-2026.63 (543.57)	-1872.87 (554.32)	0.1709 (0.1338)	0.0489 (0.0262)	-1485.85 (1105.95)
VII	173.18 (14.92)	-2104.19 (613.75)	-1954.53 (609.65)	0.2015 (0.1448)	0.0600 (0.0404)	-1701.62 (1206.13)
I to VII	88.06 (9.99)	-1426.92 (267.60)	-1286.44 (259.29)	0.1448 (0.1219)	0.0484 (0.0256)	-848.38 (658.25)
Baseline Mean	267.78	8987.63	10515.39	1.0704	0.1283	21498.74

SDID Estimates by Policy Phase: Non-Balance-Matchers

Phase	Minimum Payment (\$) (1)	Revolving Debt (\$) (2)	Statement Balance (\$) (3)	Any Delinquency Transition (pps) (4)	Any Default Transition (pps) (5)	Credit Limit (\$) (6)
I	45.09 (14.10)	-710.90 (130.92)	-625.90 (96.80)	0.2313 (0.0899)	0.0193 (0.0487)	-316.81 (464.60)
II	41.65 (15.64)	-1081.71 (144.11)	-944.23 (171.42)	0.0858 (0.1235)	-0.0184 (0.0368)	-538.55 (604.39)
III	55.85 (12.60)	-1124.26 (209.89)	-992.41 (191.76)	0.0495 (0.1038)	-0.0315 (0.0265)	-791.62 (662.23)
IV	89.98 (9.15)	-1406.47 (326.02)	-1261.70 (339.83)	0.0841 (0.1082)	-0.0333 (0.0277)	-1031.89 (912.78)
V	130.80 (12.44)	-1613.65 (530.47)	-1481.32 (549.64)	0.1484 (0.1371)	0.0028 (0.0383)	-1301.59 (1172.11)
VI	167.87 (18.70)	-1864.56 (662.27)	-1741.14 (673.04)	0.2243 (0.1507)	0.0144 (0.0434)	-1727.66 (1397.39)
VII	187.00 (21.33)	-1965.14 (722.07)	-1826.01 (706.09)	0.2258 (0.1650)	0.0186 (0.0543)	-2045.20 (1402.79)
I to VII	97.26 (8.97)	-1359.17 (319.06)	-1232.19 (313.79)	0.1451 (0.1158)	-0.0055 (0.0276)	-1048.28 (910.07)
Baseline Mean	255.53	8324.91	10042.76	2.1333	0.2991	23890.35

SDID Estimates By Credit Score for Phase VII

A. Minimum Payment (\$)

Phase	Subprime (1)	Near Prime (2)	Prime (3)	Prime Plus (4)	Superprime (5)
VII	137.21 (50.52)	126.46 (12.16)	108.86 (6.43)	78.54 (4.43)	59.78 (2.82)
Baseline Mean	234.49	171.31	165.35	113.12	71.81

B. Revolving Debt (\$)

Phase	Subprime (1)	Near Prime (2)	Prime (3)	Prime Plus (4)	Superprime (5)
VII	-278.42 (367.29)	-1339.17 (362.82)	-1499.19 (412.53)	-745.64 (277.97)	-83.39 (105.20)
Baseline Mean	4490.06	4406.92	4714.14	2393.10	807.95

C. Credit Limit (\$)

Phase	Subprime (1)	Near Prime (2)	Prime (3)	Prime Plus (4)	Superprime (5)
VII	-3061.85 (1208.25)	-2044.28 (1167.60)	-3041.58 (1292.49)	-2501.75 (1438.41)	-987.57 (1039.82)
Baseline Mean	7420.64	9747.64	12410.08	13110.59	14435.21

SDID Estimates On Credit Card Utilization for Phase VII

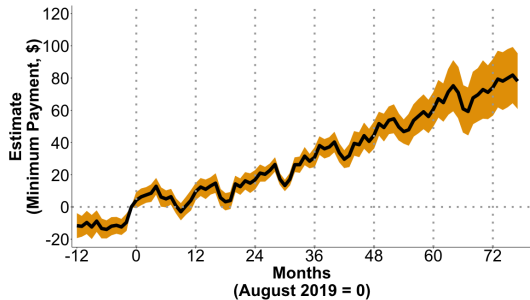
Phase	All (1)	Revolver (2)	Transactor (3)
VII	0.39 (0.76)	1.32 (1.16)	-0.07 (0.78)
Baseline Mean	29.82	35.20	10.64

Phase	Subprime (1)	Near Prime (2)	Prime (3)	Prime Plus (4)	Superprime (5)
VII	7.25 (3.02)	-2.01 (1.24)	-2.53 (1.04)	-1.03 (0.66)	0.25 (0.25)
Baseline Mean	65.42	57.60	47.55	26.29	11.89

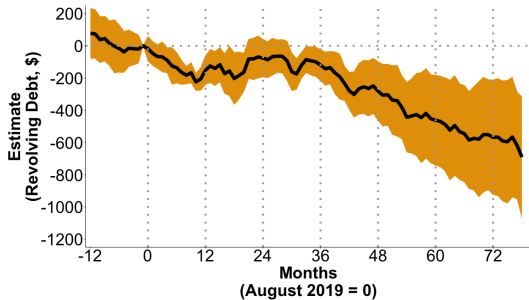
Phase	Exact Min. Payer (1)	Near Min. Payer (2)	Mixed Payer (3)	Full Payer (4)
VII	-3.56 (1.49)	-4.08 (0.98)	-0.13 (2.38)	-1.47 (0.82)
Baseline Mean	66.78	69.22	56.60	18.57

DID: Minimum Payments and Revolving Debt 🚀

A. Minimum Payment (\$)

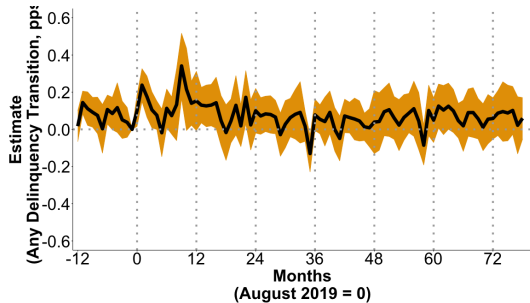


B. Revolving Debt (\$)

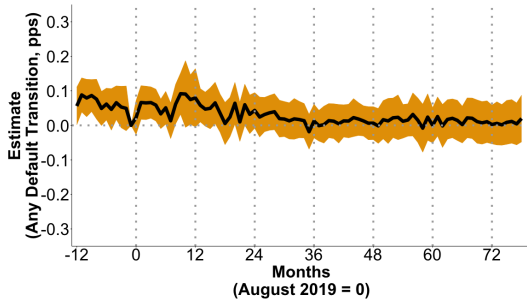


DID: Delinquency & Default Transitions

A. Any Delinquency Transition



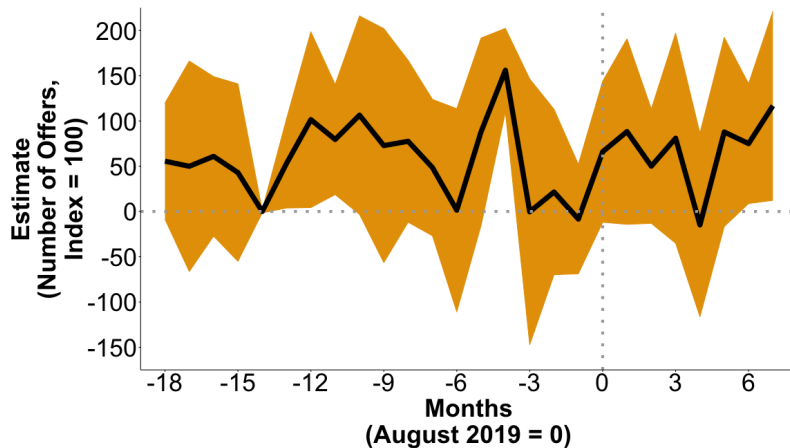
B. Any Default Transition



DID Estimates By Policy Phase

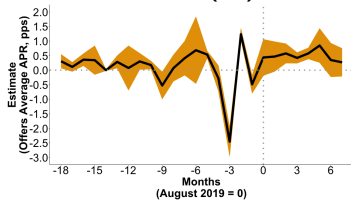
Phase	Minimum Payment (\$) (1)	Revolving Debt (\$) (2)	Statement Balance (\$) (3)	Any Delinquency Transition (pps) (4)	Any Default Transition (pps) (5)	Credit Limit (\$) (6)
I	14.90 (1.68)	-134.93 (26.75)	-87.33 (18.29)	0.0751 (0.0330)	0.0023 (0.0136)	-216.55 (117.02)
II	20.92 (2.90)	-147.34 (39.71)	-58.12 (58.31)	0.0259 (0.0385)	-0.0113 (0.0187)	-117.13 (171.25)
III	32.23 (2.80)	-108.98 (62.75)	-69.04 (64.89)	-0.0220 (0.0288)	-0.0340 (0.0183)	-178.45 (215.83)
IV	46.65 (3.73)	-227.22 (106.12)	-231.60 (134.56)	-0.0260 (0.0250)	-0.0445 (0.0106)	-449.14 (404.04)
V	61.89 (4.94)	-391.19 (154.86)	-417.24 (187.79)	-0.0081 (0.0486)	-0.0412 (0.0154)	-852.49 (613.29)
VI	77.56 (7.58)	-539.56 (197.26)	-576.57 (238.16)	0.0073 (0.0460)	-0.0428 (0.0171)	-1378.09 (865.23)
VII	84.67 (8.79)	-609.13 (222.89)	-650.39 (258.47)	0.0079 (0.0522)	-0.0475 (0.0190)	-1610.59 (951.03)
I to VII	44.25 (3.22)	-278.28 (92.46)	-262.27 (101.27)	0.0110 (0.0299)	-0.0287 (0.0136)	-595.59 (421.56)

Mintel Data: Number (Indexed, August 2018 = 100) 🚀

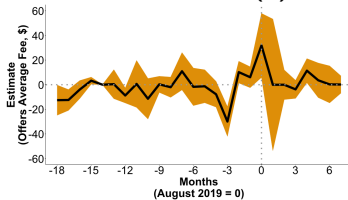


Mintel Data: Pre-Policy Offers have More Favorable Contract Terms 📍

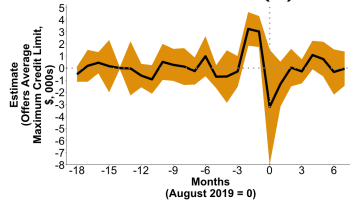
A. APR (%)



B. Annual Fee (\$)



C. Credit Limit (\$)



SDID Estimates By Policy Phase: New Card Openings

Phase	Number of New Credit Cards Opened Per 100 Consumers (1)	Value (\$) of New Credit Card Limits Per 100 Consumers (2)	Mean Credit Card Limit (\$) of New Cards Opened (\$) (3)	Mean Credit Score of New Cards Opened (4)
I	-0.136 (0.081)	-342 (376)	-51 (139)	0.46 (4.68)
II	-0.157 (0.065)	81 (392)	378 (373)	1.85 (6.68)
III	-0.344 (0.023)	-808 (392)	417 (309)	1.60 (6.28)
IV	-0.415 (0.040)	-977 (358)	628 (241)	4.59 (4.48)
V	-0.351 (0.050)	-771 (370)	538 (222)	5.53 (3.78)
VI	-0.287 (0.070)	-578 (414)	394 (353)	7.12 (3.60)
VII	-0.255 (0.082)	-1589 (591)	-963 (502)	10.50 (5.68)
I to VII	-0.279 (0.052)	-657 (340)	265 (276)	4.14 (4.73)
Baseline Mean	1.779	7038	3956	771.66

Model of Issuers: Environment & Borrower Behavior

J banks & continuum of borrowers with types $\theta \in [0, 1]$, where higher θ is more creditworthy

Bank customer base from distribution $F_j(\theta)$

Three periods, $t \in \{1, 2, 3\}$. Assume in period 1 all have same formula. In period 2 policy requires 2%, and in period 3 5%. Policy announced before period 2 and is known to all.

Borrower: of type θ with credit limit L and minimum payment rate m has revolving debt:

$$D(\theta, m, L) = \alpha(\theta)(1 - m)L, \quad (8)$$

$\alpha(\theta) \in [0, 1]$ is the utilization rate, with $\alpha'(\theta) < 0$. Transactors have $\alpha(\theta) \approx 0$.

Default probability:

$$p(\theta, m) = \bar{p}(\theta) + \chi \cdot H(m - \psi(\theta)), \quad H(x) \equiv \mathbb{E}_\varepsilon[(x - \varepsilon)^+], \quad (9)$$

$\bar{p}(\theta)$ is the baseline default probability ($\bar{p}'(\theta) < 0$)

$\psi(\theta)$ is the maximum share of the balance that type θ can afford to repay ($\psi'(\theta) > 0$)

$\chi > 0$ governs default sensitivity to liquidity stress, and ε is a mean-zero shock

Model of Issuers: Bank Profit and Optimal Credit Limits

Bank j earns per-period profit from borrower θ with credit limit L :

$$\pi(\theta, m, L) = \underbrace{\pi_0 \cdot L}_{\text{Interchange revenue}} + \underbrace{(r - c) \cdot \alpha(\theta)(1 - m)L}_{\text{Net interest margin}} - \underbrace{p(\theta, m) \cdot \lambda L}_{\text{Expected default losses}} \quad (10)$$

$\pi_0 > 0$ per-dollar interchange, r interest rate, c cost of funds, and $\lambda \in (0, 1)$ loss given default
Bank chooses credit limit L to maximize $\pi(\theta, m, L) - (\kappa/2)L^2$, where $(\kappa/2)L^2$ is convex cost
Define the net margin per dollar of credit extended as:

$$N(\theta, m) \equiv \pi_0 + (r - c)\alpha(\theta)(1 - m) - p(\theta, m)\lambda. \quad (11)$$

Then $\pi(\theta, m, L) = N(\theta, m) \cdot L$ and the F.O.C. w.r.t. L yields:

$$L_j^*(\theta, m) = \frac{N(\theta, m)}{\kappa}, \quad v(\theta, m) \equiv \frac{[N(\theta, m)]^2}{2\kappa} \quad (12)$$

Bank serves θ iff $N(\theta, m) \geq 0$ so types $\theta < \bar{\theta}(m)$ are denied credit & transactors always served

Bank j aggregate profit: $\Pi_j(m) = \int_{\bar{\theta}(m)}^1 \frac{[N(\theta, m)]^2}{2\kappa} dF_j(\theta)$

Status quo (low) minimum payment m^{SQ} is profit-maximizing: $\Pi_j(m^{SQ}) \geq \Pi_j(m), \forall m > m^{SQ}$

Model Result 1: Credit Limit Reduction 🦋

Regulation forces m above m^{SQ} at $t = 2$ causing credit supply contracting via 2 channels:

1. Credit rationing.

Totally differentiating $N(\bar{\theta}, m)$ gives $d\bar{\theta}/dm = -(\partial N/\partial m)/(\partial N/\partial \theta)$

denominator is positive because the default-reduction dominates lost-interest as θ rises, otherwise the cutoff is not well-defined

Numerator is negative as N falls with m both via lost interest margin and via increased default risk for borrowers near payment capacity

Hence, $d\bar{\theta}/dm > 0$

2. General decline in credit limits.

From eq. (12), $\partial L_j^*/\partial m < 0$

Credit limits fall for all revolvers when minimum payments rise above the status quo

Assumption 1: motivated by bank lending officers, who claim substantial fixed costs

- Each time bank j changes its minimum payment formula, it incurs a fixed cost $K_j > 0$
- Cost varies across banks to capture differences in IT modifications, communications etc.

Model Result 2: Over-Compliance 📌

At $t = 2$, the regulation forces issuers using the status quo formula to switch to a percentage-of-balance rule in Quebec and floor will tighten to $\bar{m}_T = 5\%$ at $t = 3$

The bank's problem at $t = 2$ in Quebec is:

$$V_j^{QC} = \max_{m_j \in \{\bar{m}_1, \bar{m}_T\}} \left\{ \Pi_j(m_j) - K_j + \beta_B [\Pi_j(\bar{m}_T) - K_j \cdot \mathbf{1}\{m_j \neq \bar{m}_T\}] \right\}. \quad (13)$$

At $t=2$, comparing minimal compliance (strategy A) to over-compliance (strategy B) yields:

$$V_j^B - V_j^A = -[\Pi_j(\bar{m}_1) - \Pi_j(\bar{m}_T)] + \beta_B K_j. \quad (14)$$

At $t = 2$, bank j jumps to $\bar{m}_T$ in Quebec if and only if the discounted menu cost it saves at $t = 3$ exceeds the profit forgone from setting a rate above the current floor at $t = 2$:

$$\beta_B K_j > \Pi_j(\bar{m}_1) - \Pi_j(\bar{m}_T), \quad (15)$$

Can be because K_j is high or the profit gap is small (transactor-heavy portfolios)

Model Result 3: No Defaults 🦋

Regulation has two opposing effects on equilibrium default:

1. Every served borrower faces a higher default probability ($\partial p / \partial m > 0$)
2. The cutoff $\bar{\theta}(m)$ rises so the highest-risk borrowers are rationed out

Net sign depends on F , the curvature of H , and the variance of ε

Empirically, no change in defaults is consistent with the two margins offsetting

Mean portfolio of a credit card issuer by their compliance status in Phase I

Over-compliers have a higher transactor share and lower revolving debt

→ consistent with the composition channel

	Just-comply (1)	Over-comply (2)	Already-comply (3)
Mean Transactors (%)	42.2	48.3	43.2
Transactors (mn)	0.175	0.303	0.127
Mean revolving balance (\$)	1,792	1,379	1,040
Revolving balances (\$ mn)	990	1,101	544
Mean statement balance (\$)	2,354	2,037	1,551
Statement balances (\$ mn)	1,280	1,643	1,764
Accounts (mn)	0.575	0.846	1.163

– Few issuers (2 just-comply; 5 over-comply; 4 already-comply)
so the composition channel cannot be cleanly separated from issuer size

Sensitivity of Break-Even To Payer Types Given Non-Zero Internality Benefit

	No supply correction			External			Model-implied		
$\beta :$	0.55	0.74	0.83	0.55	0.74	0.83	0.55	0.74	0.83
Near Min. Payers	36	23	14	21	13	9	16	9	6
+ Mixed Payers	148	85	56	54	31	20	43	25	16
+ Full Payers	259	150	98	107	62	41	106	61	40
+ Exact Min. Payers	270	156	102	114	66	43	110	64	42

- Use Phase-I debt response to identify β and then apply to Phase-VII debt response
- Phase-I used because $\Delta L \approx 0$

	ΔD^{ident} (\$)	ΔD^{VII} (\$)	$\hat{\beta}$	$\hat{\gamma}$ (per \$)	Supply share
<i>Baseline (supply-unadjusted; Phase VII identifies and scales)</i>					
Near Min. Payer	2,089	2,089	0.75	0.25	—
Mixed Payer	1,107	1,107	0.86	0.14	—
<i>Correction 1 (Phase I clean window)</i>					
Near Min. Payer	122	2,089	0.98	0.02	limit rose
Mixed Payer	396	1,107	0.95	0.05	n/a
<i>Correction 2a (model-implied MPB)</i>					
Near Min. Payer	937	2,089	0.88	0.12	55%
Mixed Payer	266	1,107	0.96	0.04	76%

Supply-Adjusted Break-Even Access Cost

Calibration	Λ^{be} (\$)	% of baseline
Baseline (supply-unadjusted)	54	100%
Correction 1 (Phase I clean window)	14	26%
Correction 2 (supply-driven-response subtraction)		
2a: model-implied	18	34%
2b: external MPB	23	43%